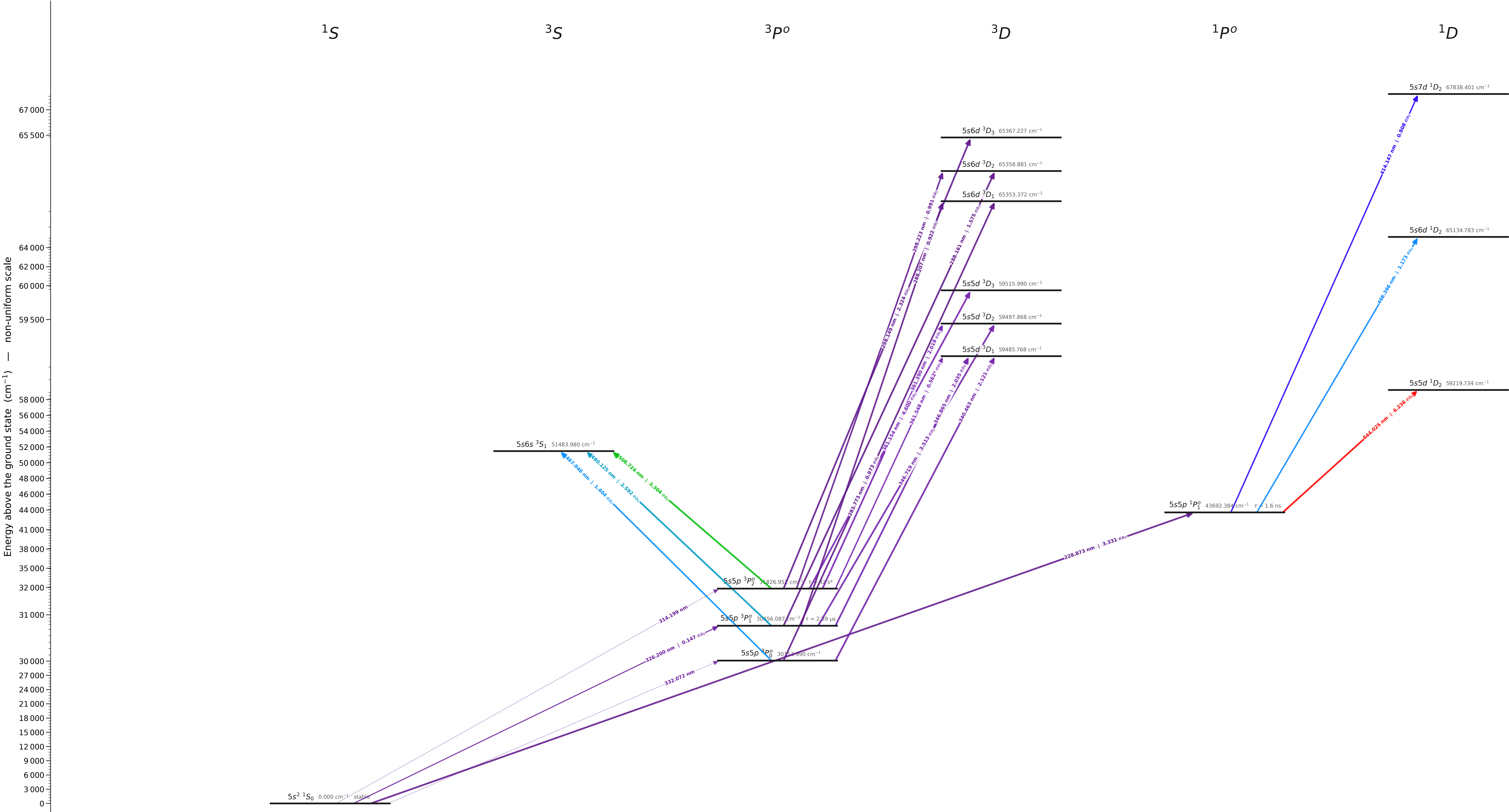
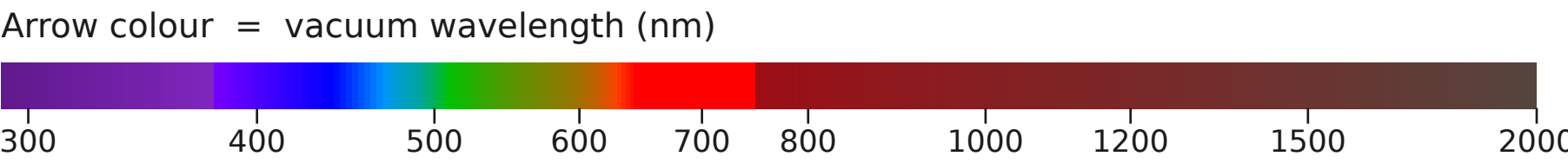


Cadmium-113

¹¹³Cd (Cd I), nuclear spin *I* = 1/2. Strongest classified transitions below 2 μm (NIST): 15 levels, 21 lines.

Cd⁺ ionisation limit 72540.05 cm⁻¹ = 8.99382 eV (137.855 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element $|U\|e r \|U'\rangle|$ in ea0
Matrix elements are measured or NIST values; * = high-accuracy theory, ≈ = model or semi-empirical calculation.
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, *n* = 70.
Level caption: energy (NIST, or a measurement of the last 20 years where one exists) and measured lifetime (where one exists).



Key transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	$ U\ e r \ U'\rangle $ (ea0)	A (s ⁻¹)	Γ/2π up. (MHz)	branch (%)	Use
5s ² 1S ₀ - 5s5p 1P ₁ ^o	228.8728	228.8023	1309.8647	3.331	6.250e+08	99.47	100	228.9 nm resonance line: first-stage MOT / slowing / imag...
5s ² 1S ₀ - 5s5p 3P ₂ ^o	314.1991	314.1081	954.1480	-	-	3.789e-09	-	314 nm ultra-narrow
5s ² 1S ₀ - 5s5p 3P ₁ ^o	326.1995	326.1055	919.0464	0.147	4.184e+05	0.06659	100	326.1 nm intercombination line: narrow-line MOT
5s ² 1S ₀ - 5s5p 3P ₀ ^o	332.0716	331.9760	902.7947	-	-	-	-	332 nm optical lattice clock transition
5s5p 3P ₁ ^o - 5s5d 3D ₃	361.1538	361.0508	830.0965	4.600	1.300e+08	-	-	361.2 nm metastable MOT
5s5p 3P ₂ ^o - 5s5d 3D ₁	361.5484	361.4453	829.1904	0.562*	4.510e+06	-	-	361.5 nm
5s5p 3P ₁ ^o - 5s6s 3S ₁	480.1254	479.9912	624.4045	2.592	4.100e+07	-	-	480.0 nm optical pumping / repumping

λ, ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.
Γ/2π: natural linewidth of the upper level, 1/(2πτ).

¹¹³Cd hyperfine structure (*I* = 1/2)

Level	A (MHz)	B (MHz)	Shift of each F level from the centre of gravity (MHz)
5s5p 3P ₁ ^o	-4313.860(13)	-	F=1/2: +4313.86 F=3/2: -2156.93
5s5p 3P ₂ ^o	-3444.6344(16)	-	F=3/2: +5166.95 F=5/2: -3444.63
5s5p 1P ₁ ^o	197.1(23)	-	F=1/2: -197.10 F=3/2: +98.55
5s6s 3S ₁	-8108.6(9)	-	F=1/2: +8108.60 F=3/2: -4054.30
5s5d 3D ₃	-2496.251(26)	-	F=5/2: +4992.50 F=7/2: -3744.38

Measured A, B constants; sources in the data file.

Isotope shifts

Transition	λ vac (nm)	Isotopes	Shift (MHz)
5s ² 1S ₀ - 5s5p 1P ₁ ^o	228.873	106-114	1818(4)
5s ² 1S ₀ - 5s5p 1P ₁ ^o	228.873	108-114	1336(3)
5s ² 1S ₀ - 5s5p 1P ₁ ^o	228.873	110-114	865(3)
5s ² 1S ₀ - 5s5p 1P ₁ ^o	228.873	112-114	408(3)
5s ² 1S ₀ - 5s5p 1P ₁ ^o	228.873	116-114	-316(3)
5s ² 1S ₀ - 5s5p 1P ₁ ^o	228.873	111-114	805.0(3)
5s ² 1S ₀ - 5s5p 1P ₁ ^o	228.873	113-114	344.9(5)
5s ² 1S ₀ - 5s5p 3P ₂ ^o	314.199	106-114	1916(4)
5s ² 1S ₀ - 5s5p 3P ₂ ^o	314.199	108-114	1404(3)
5s ² 1S ₀ - 5s5p 3P ₂ ^o	314.199	110-114	913.7(21)
5s ² 1S ₀ - 5s5p 3P ₂ ^o	314.199	112-114	429.7(16)
5s ² 1S ₀ - 5s5p 3P ₂ ^o	314.199	116-114	-320.2(17)
5s ² 1S ₀ - 5s5p 3P ₁ ^o	326.200	106-114	1913.0(10)
5s ² 1S ₀ - 5s5p 3P ₁ ^o	326.200	108-114	1402.4(10)
5s ² 1S ₀ - 5s5p 3P ₁ ^o	326.200	110-114	914.7(10)
5s ² 1S ₀ - 5s5p 3P ₁ ^o	326.200	112-114	429.9(10)
5s ² 1S ₀ - 5s5p 3P ₁ ^o	326.200	116-114	-321.5(10)
5s ² 1S ₀ - 5s5p 3P ₁ ^o	326.200	111-114	861.6(10)
5s ² 1S ₀ - 5s5p 3P ₁ ^o	326.200	113-114	370.8(10)
5s ² 1S ₀ - 5s5p 3P ₀ ^o	332.072	106-114	1911(5)
5s ² 1S ₀ - 5s5p 3P ₀ ^o	332.072	108-114	1404(4)
5s ² 1S ₀ - 5s5p 3P ₀ ^o	332.072	110-114	914.1(21)

Shift = ν(first isotope) - ν(second isotope).